

Auto Loan Securitisation Risk Analytics

A Power BI and DAX analytical framework for portfolio performance, delinquency migration, vintage analysis, IFRS 9 ECL and stress testing.

A consolidated reference report accompanying a ten-page interactive dashboard built on a 500-loan illustrative auto-loan ABS dataset. The framework is structured around the Day 7 – Day 11 workflow specified in the securitisation project manual: portfolio composition, days-past-due migration, static-pool vintages, IFRS 9 expected credit loss, parameterised stress, and investor reporting.

500	₹546.07M	₹317.79M	₹11.66M
LOANS IN POOL	ORIGINAL POOL VALUE	OUTSTANDING BALANCE	NET IFRS 9 ECL (OVERVIEW)

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01. Executive Summary

This report consolidates a Power BI and DAX analytical framework built on a 500-loan illustrative auto-loan securitisation pool. It translates raw, multi-source loan-level data into investor-grade risk intelligence covering portfolio composition, delinquency migration, vintage performance, IFRS 9 expected credit loss, and parameterised stress testing.

HEADLINE SNAPSHOT

- 500 active loan records, original pool value ₹546.07 M, outstanding ₹317.79 M
- Weighted average coupon 10.9 %, pool factor 0.58, cumulative recovery rate 31 %
- Net IFRS 9 ECL of ₹11.66 M at the overview grain; ₹59.14 M lifetime ECL at the stage-clause grain
- Default rate 1.80 %, maximum DPD 694, average DPD 21.21

The central contribution of this project is the linkage between operational portfolio monitoring and forward-looking credit-risk analysis. Days-past-due migration supplies the early-warning layer; roll-rate transition probability shows how borrowers cross delinquency thresholds; vintage curves show how cohorts deteriorate at comparable seasoning; the IFRS 9 framework translates probability of default, loss given default and exposure at default into expected credit loss; and the What-if stress engine converts adverse macro assumptions into stressed ECL.

Metric	Value	Dashboard Page
Active loans (current pool view)	476 of 500	Portfolio Analytics
Original pool value	₹546.07 M	Executive Overview
Outstanding balance	₹317.79 M	Executive Overview
Average CIBIL / DTI	739.24 / 0.24	Portfolio Analytics
90+ DPD rate	0.07 (7 %)	Credit Risk & IFRS 9
Default rate	1.80 %	Executive Overview
Recovery rate (cumulative)	31 %	Executive Overview
WAC / pool factor	10.9 % / 0.58	Executive Overview
Total EAD	₹326.16 M	Credit Risk & IFRS 9
IFRS 9 ECL (lifetime, stage view)	₹59.14 M	IFRS 9 ECL Analysis
Net ECL (overview snapshot)	₹11.66 M	Executive Overview
Average DPD	21.21 days	Portfolio Performance
Maximum DPD	694 days	Portfolio Performance

The dashboard suite is delivered as ten paginated pages. Pages one through six are decision-grade — Executive Overview, Portfolio Analytics, Credit Risk & IFRS 9, Portfolio Performance, Investor & Stress Analytics, and Loan-Level Drill-Through. Pages seven through ten are analytical — DPD & Delinquency, Vintage / Static-Pool, IFRS 9 ECL Analysis, and Stress Testing. Each page is structured around KPI tiles, distribution charts, evolution charts, and (where relevant) parameterised sliders that drive the underlying DAX measures.

02. Project Objective & Scope

The objective of this project is to transform raw securitisation data into a repeatable analytical framework that supports credit committees, risk teams, investors, servicers and regulators. The analytical model is anchored in Power BI Desktop, organised around a dimensional star schema, and operationalised through 30+ foundational DAX measures, an additional layer of advanced iterator / virtual-table measures, and a what-if parameter layer for stress testing.

Scope Boundaries

- **Portfolio health.** Pool balance, loan count, outstanding exposure, pool factor and weighted borrower / loan metrics.
- **Credit performance.** Delinquency buckets, 90+ DPD, default rate, roll-rate transitions, ageing.
- **Static-pool analysis.** Cumulative default, cumulative net loss and cohort-level loss trajectories by vintage.
- **Expected loss.** IFRS 9 stage classification, PD, LGD, EAD and ECL.
- **Scenario analysis.** Default-rate shock, recovery-rate haircut, prepayment shock and interest-rate shock.
- **Investor view.** Pool balance, prepayments, recoveries, gross loss, net loss.
- **Drill-through.** Loan-level borrower, balance, CIBIL, DPD and status information.

ILLUSTRATIVE — NOT AN INSTITUTIONAL PORTFOLIO

The source dataset contains approximately 500 loans and is supplied for academic / framework-construction purposes. All KPIs are reproduced from the supplied dashboard screenshots and source datasets. Where different pages use different analytical grains — current pool versus static-pool history — figures are reported in their page-specific context. This is a stress-testing and IFRS 9 framework implementation in Power BI, not a regulatory ECL model.

03. Securitisation Domain Overview

Securitisation aggregates receivables into a pool that supports securities issued to investors. For an auto-loan pool, the underlying receivables are the contractual monthly payments of borrowers. Portfolio-level risk therefore begins at the individual loan and propagates through delinquency, default, loss, recovery and cash-flow behaviour until the pool amortises or restructures.



Instrument taxonomy in scope

- **Pass-through certificates.** Cash flows from the underlying pool pass through to investors proportionally.
- **Pay-through securities (CMOs).** Cash flows are restructured into sequential tranches with different maturities.
- **Credit-tranched securities (CDOs/CLOs).** Cash flows are allocated by seniority — senior tranches absorb losses last; equity tranches absorb losses first.

The project framework is built around early-warning and forward-looking measures rather than relying on default counts alone. The Day 7 – Day 11 layers combine DPD transition analysis, vintage loss curves, IFRS 9 ECL, and parameterised stress testing.

04. Dataset & Data Architecture

The manual identifies four core CSV datasets. These were mapped to rating-agency requirement tables on Day 1 of the methodology, and the same field universe was subsequently rewritten into a Power BI dimensional model.

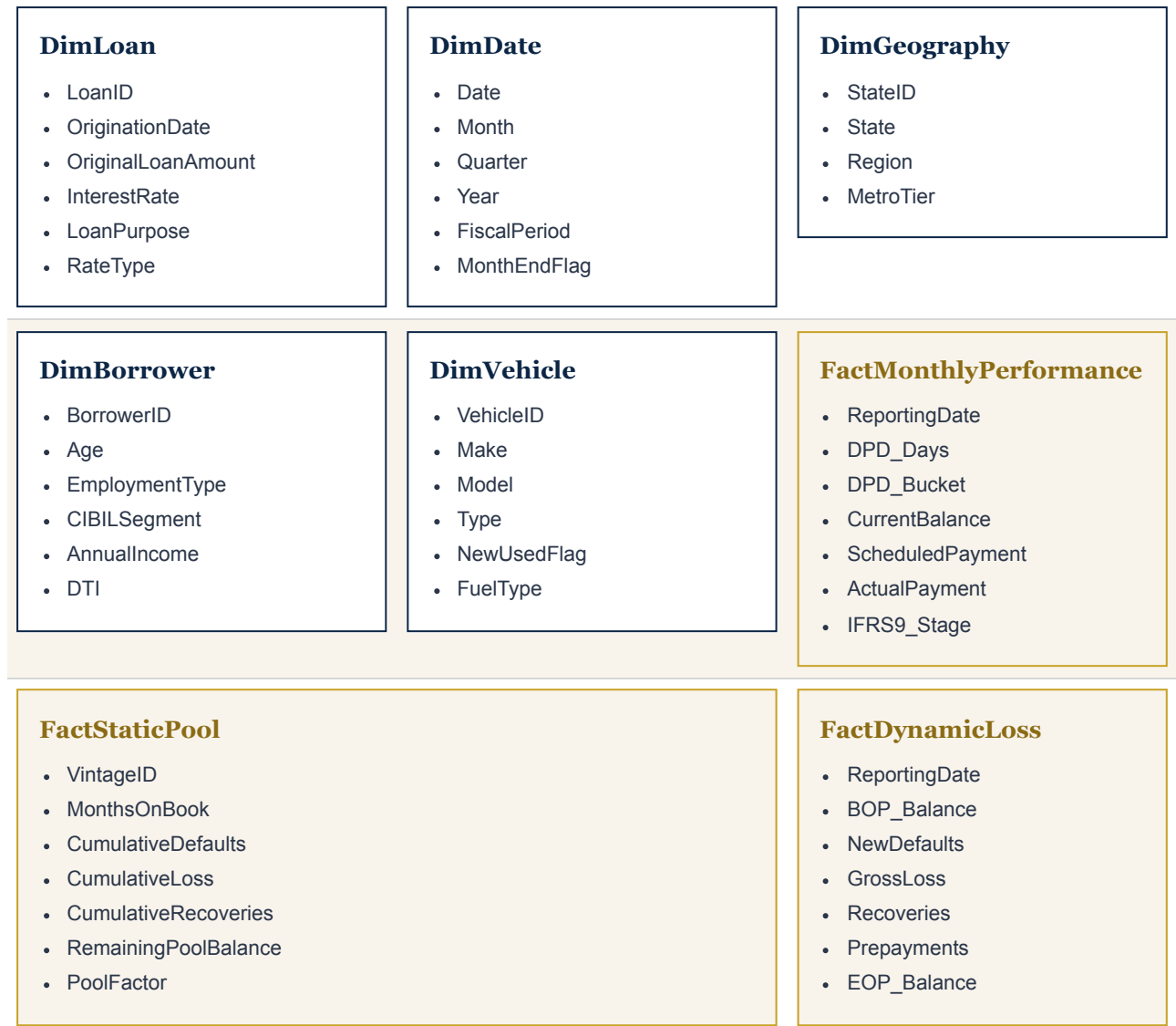
Source Dataset	Approx. Rows	Grain	Purpose
auto_loan_securitisation_data.csv	500	One row per loan	Loan attributes, balances, DPD / status, IFRS 9 fields, losses / recoveries
dpd_snapshot_history.csv	6,000	Loan × monthly snapshot	Delinquency migration, ageing, roll-rate matrix
static_pool_vintage_data.csv	375	Vintage × Months-on-Book	Cumulative default / loss curves and pool-factor run-off
dynamic_loss_monthly.csv	12	Reporting month	Defaults, losses, recoveries, collections, prepayments, end-of-period balance

Field universe (loan-level table)

- **Identification.** Loan ID, Pool ID, Originator ID, Servicer ID, Account Status.
- **Origination.** Origination Date, Original Balance, Original Term, Interest Rate, Rate Type, Origination Channel, Loan Purpose.
- **Vehicle / collateral.** Vehicle Make, Model, Year, Type, New/Used flag, Original Vehicle Value, Current Vehicle Value, Mileage at Origination, Registration State.
- **Borrower.** CIBIL at Origination, Current CIBIL, Employment Type, Annual Income, DTI Ratio, Age, Geographic Region, Years at Current Employment.
- **Performance (monthly).** Reporting Date, Current Balance, Scheduled Payment, Actual Payment, Delinquency Status, Days Past Due, Cumulative Loss, Recovery Amount, Prepayment Amount, Modification Flag, IFRS 9 Stage.

05. Power BI Star Schema & Relationships

The Power BI semantic model follows a dimensional star-schema approach: dimensions provide descriptive context, fact tables retain transactional or periodic observations. The same measure can therefore be reused across executive, risk, vintage and investor views, with filter propagation handled by standard relationships.



Relationships are one-to-many from each dimension into the relevant fact table. Role-playing dimensions are required so that origination date drives vintage analysis while reporting date drives current-performance measures; month-end date logic is applied on the DimDate dimension.

o6. DAX & KPI Framework

The manual targets a foundational library of more than thirty DAX measures. These are grouped into eight families of metrics. Below the family map, this section reproduces the core formulas used by the dashboard pages.

Family	Representative Metrics	Business Question
Pool health	Original / Current Balance, Active Loans, Pool Factor	How much exposure remains?
Credit quality	30+, 60+, 90+ DPD rate, Default Rate	Is asset quality deteriorating?
Weighted metrics	WAC, WAM, WALA, WALT, weighted DTI	What is the risk / seasoning profile?
Loss & recovery	Gross Loss, Recoveries, Net Loss, Recovery Rate	How severe are realised losses?
Collections	Billing, Collections, Collection Efficiency	Are cash flows being collected?
Vintage	Cumulative Default / Loss Rate	Which cohorts perform worse?
ECL	PD, LGD, EAD, Stage ECL, Total ECL	What is the expected loss?
Stress	Stressed PD, stressed LGD, stressed ECL	How resilient is the portfolio?

Core DAX Measures

Pool Factor	<code>DIVIDE(SUM(Loans[CurrentBalance]), CALCULATE(SUM(Loans[OriginalLoanAmount]), ALL(DimDate)))</code>
Weighted Average Coupon (WAC)	<code>DIVIDE(SUMX(Loans, Loans[InterestRate] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]))</code>
Weighted Average Maturity (WAM)	<code>DIVIDE(SUMX(Loans, Loans[RemainingTerm] * Loans[CurrentBalance]), SUM(Loans[CurrentBalance]))</code>
Collection Efficiency	<code>DIVIDE(SUM(Loans[ActualPayment]), SUM(Loans[ScheduledPaymentDue]))</code>
30+ DPD Rate	<code>DIVIDE(CALCULATE(SUM(Loans[CurrentBalance]), Loans[DelinquencyDays] >= 30), SUM(Loans[CurrentBalance]))</code>
90+ DPD Rate (NPA)	<code>DIVIDE(CALCULATE(SUM(Loans[CurrentBalance]), Loans[DelinquencyDays] >= 90), SUM(Loans[CurrentBalance]))</code>

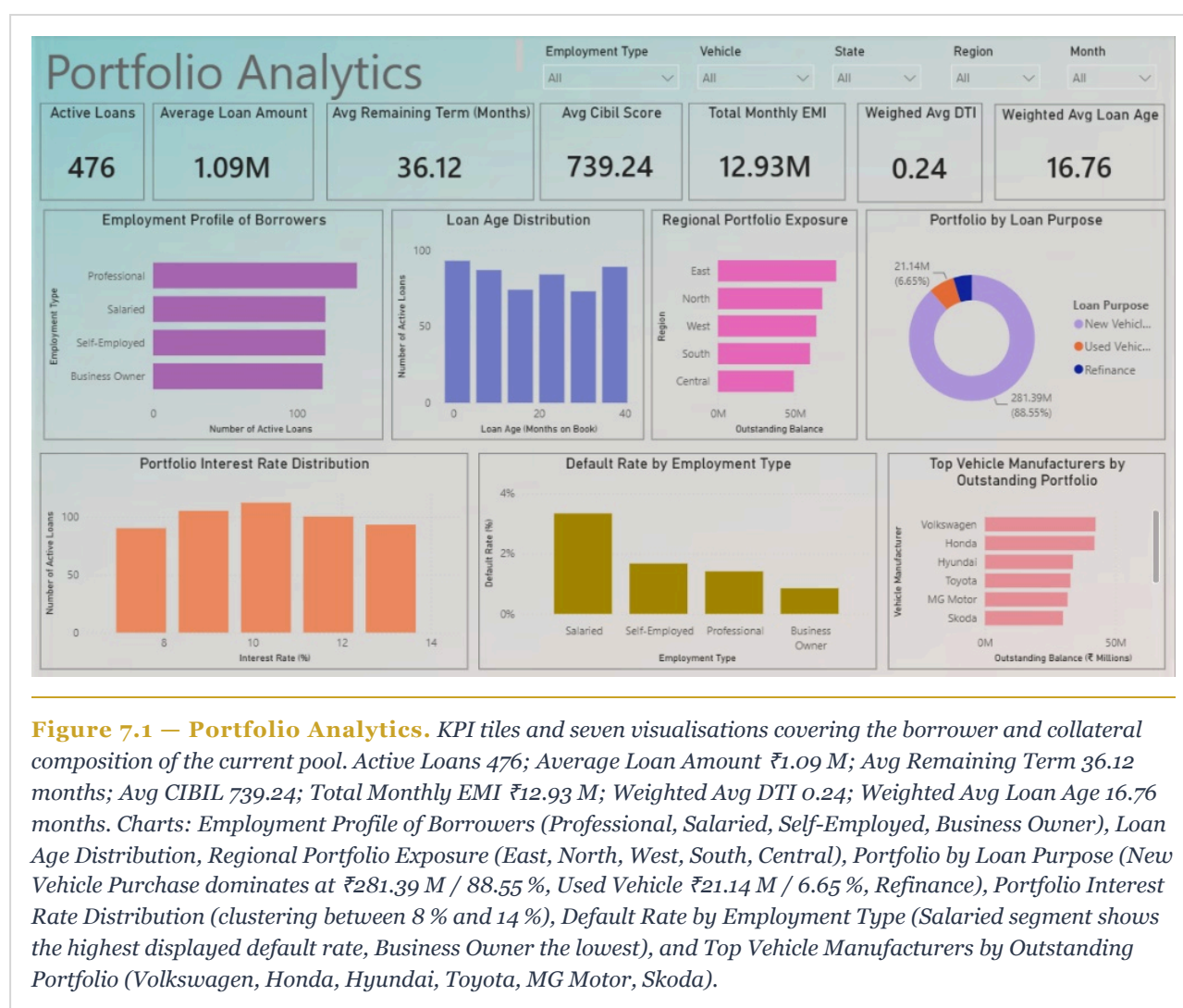
Single Month Mortality (SMM)	DIVIDE(SUM(Loans[PrepaymentAmount]), CALCULATE(SUM(Loans[CurrentBalance]), PREVIOUSMONTH(DimDate[Date])) - SUM(Loans[ScheduledPrincipal]))
Conditional Prepayment Rate (CPR)	1 - POWER(1 - [SMM], 12)
Constant Default Rate (CDR)	VAR MDR = DIVIDE([Defaults This Month], CALCULATE(SUM(Loans[CurrentBalance]), PREVIOUSMONTH(DimDate[Date]))) RETURN 1 - POWER(1 - MDR, 12)

Architecture of the What-If layer

Four what-if parameters drive the stress-testing page: Default Rate Shock, Recovery Rate Haircut, Prepayment Shock and Interest Rate Shock. The what-if layer is implemented as Power BI disconnected slicers that write to parameter measures. Downstream measures — Stressed PD, Stressed LGD, Stressed ECL — read from the parameter measures and respond live to slider changes.

07. Portfolio Analytics

The Portfolio Analytics page is the entry point for understanding pool composition. It combines KPI tiles (current portfolio shape) with seven visualisations covering borrower mix, loan-age distribution, regional exposure, loan purpose, interest-rate distribution, default-rate segmentation by employment, and top vehicle manufacturers by outstanding balance.



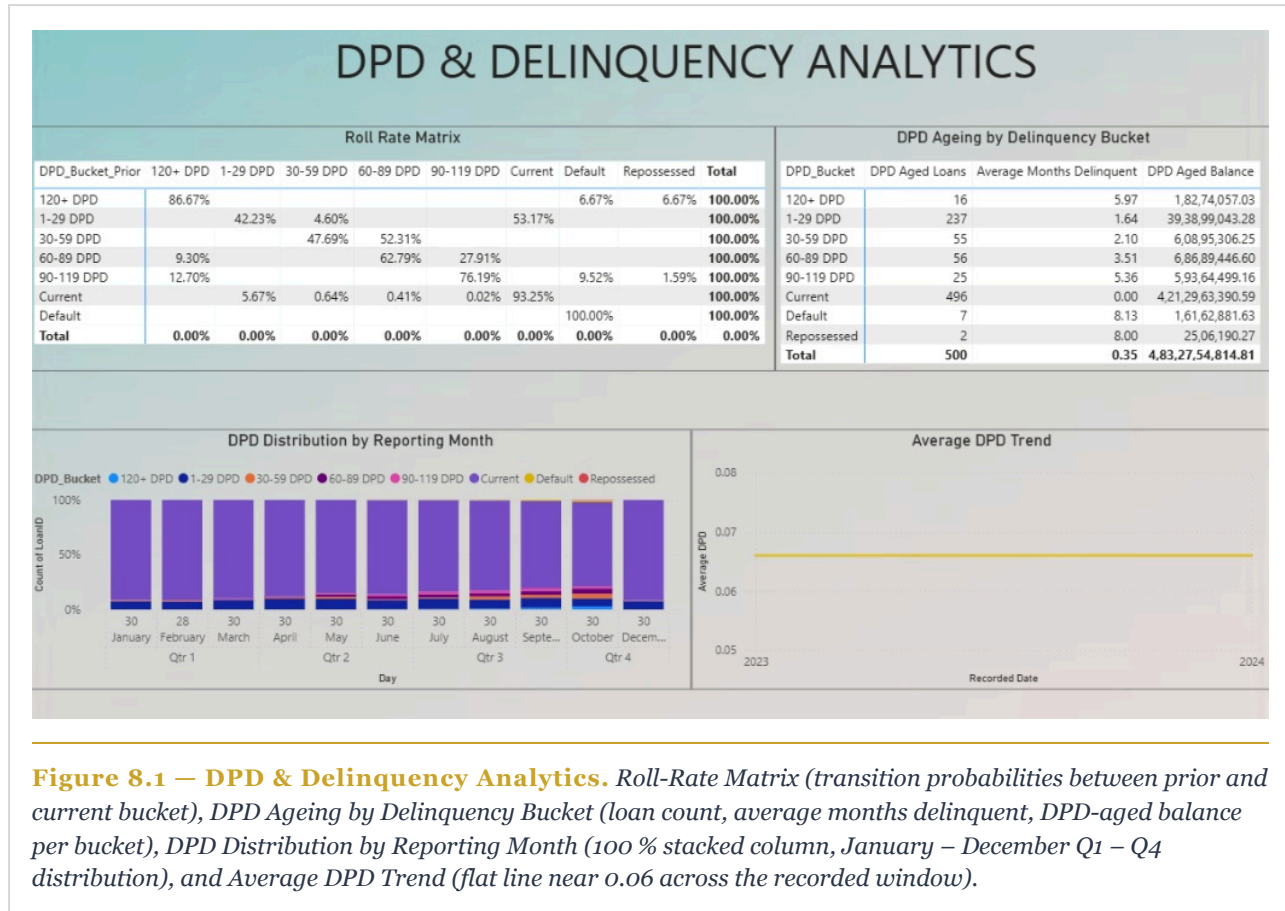
Walk-through of the seven visualisations

- **Employment profile.** Professional and Salaried segments dominate the active loan count; this concentration is material for underwriting segmentation.
- **Loan-age distribution.** Approximately uniform between 0 and 40 months-on-book, reflecting a relatively fresh book with a tail of seasoned loans.
- **Regional exposure.** East and North lead outstanding balance; concentration risk should be monitored alongside asset quality.

- **Loan purpose.** New vehicle purchase provides both collateral stability (factory warranty) and exposure to OEM / macro cycles.
- **Interest-rate distribution.** Pricing between roughly 8 % and 14 % allows rate-stress modelling.
- **Default rate by employment.** Salaried segment displays the highest default rate in the captured snapshot; Business Owner the lowest — a useful proxy for behavioural risk segmentation.
- **Top vehicle manufacturers.** Volkswagen leads the top-six outstanding-balance list; concentration should be tracked to the underlying collateral-model risk.

o8. DPD & Roll-Rate Analysis

Days-past-due (DPD) migration is the earliest quantitative signal available to a portfolio risk team. The DPD & Delinquency Analytics page implements the Day 7 deliverable: DPD distribution, roll-rate transition matrix and DPD ageing.



Read of the ageing snapshot

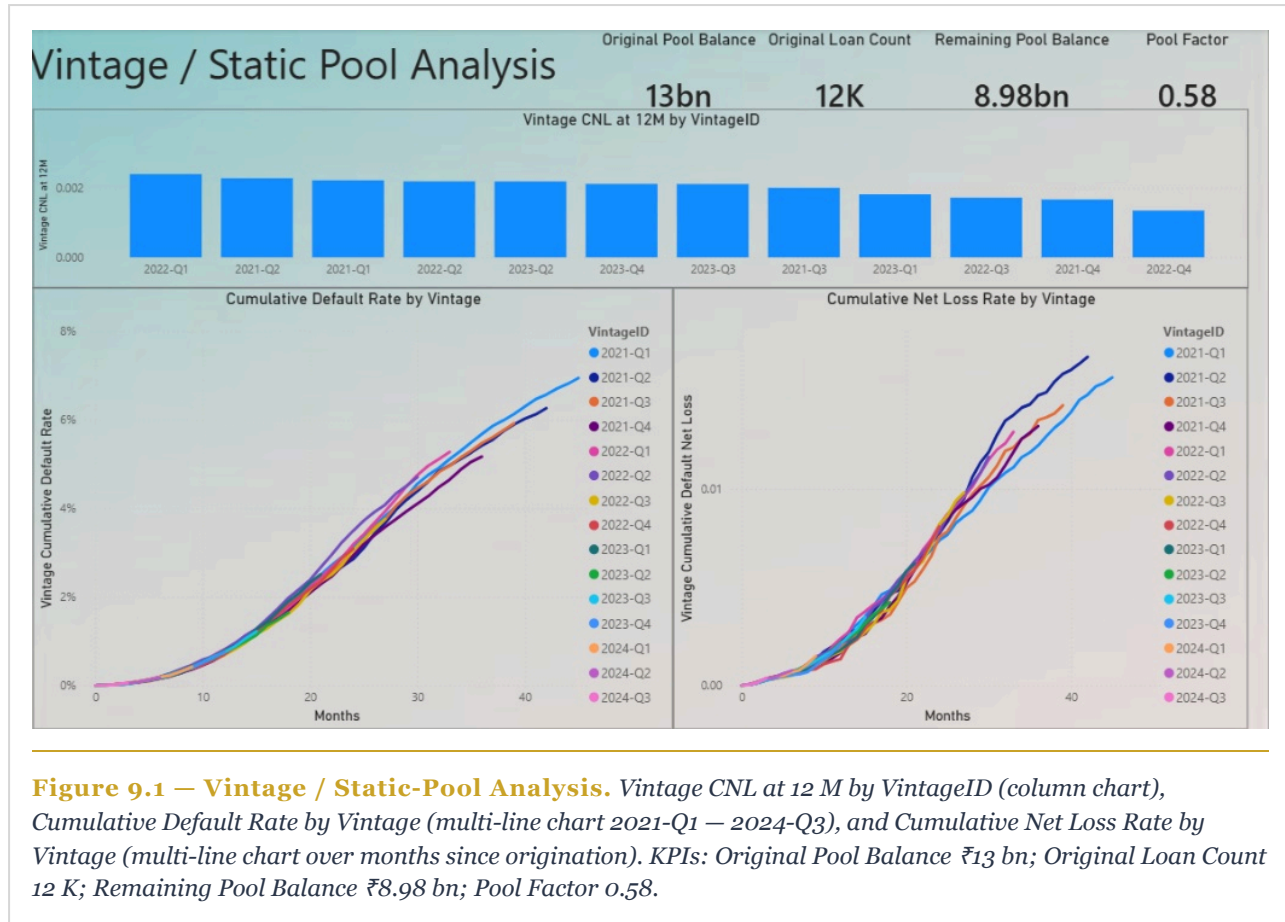
Bucket	Loan Count	Avg Months Delinquent	Bucket Aged Balance (₹)
Current	496	0.00	4,21,29,63,390
1 – 29 DPD	237	1.64	39,38,99,043
30 – 59 DPD	55	2.10	6,08,95,306
60 – 89 DPD	56	3.51	6,86,89,446
90 – 119 DPD	25	5.36	5,93,64,499
Default	7	8.13	1,61,62,881
Reposessed	2	8.00	25,06,190
Total	500	0.35	4,83,27,54,814

Early-warning logic

Default is a lagging outcome; the roll-rate matrix is the leading indicator. A spike in transitions from *Current* → *1-29 DPD* precedes a spike in *1-29* → *30-59* which in turn precedes *60-89* → *90-119* → *Default*. Monitoring the leading edge of this cascade allows the risk team to dispatch early-collection actions weeks before default materialises.

09. Vintage / Static-Pool Analysis

Vintage analysis compares origination cohorts at the same Months-on-Book (MOB). This separates cohort effects from simple portfolio ageing — a defective cohort can be identified early even when the current snapshot is dominated by healthier vintages.



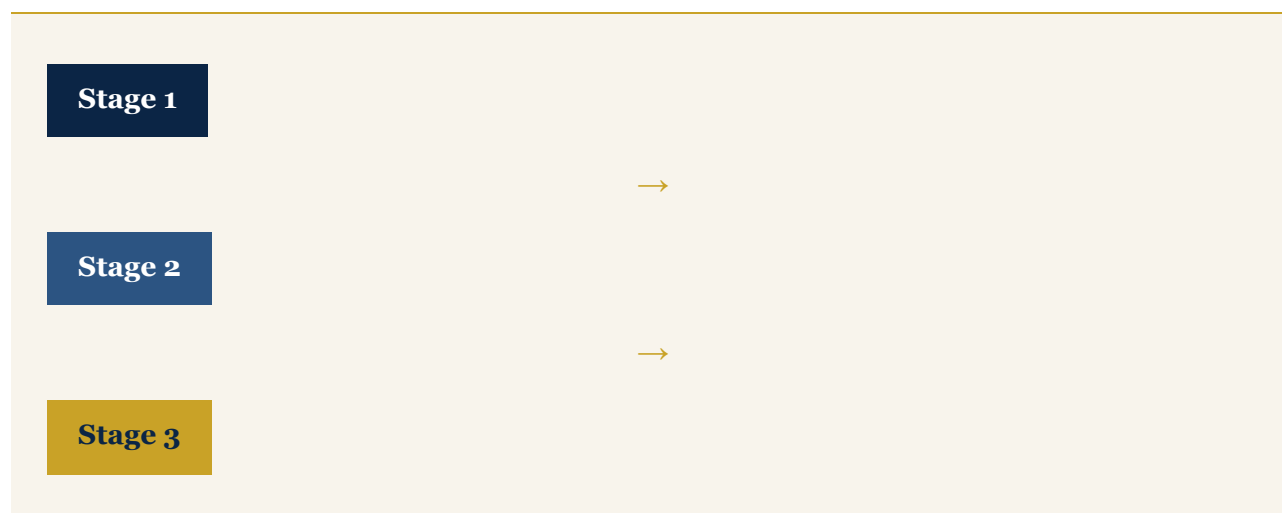
Cohort read-through

- The cumulative default rate rises monotonically with seasoning, front-loaded in the early MOB and tapering beyond 36 months.
- The cumulative net-loss rate follows the same trajectory but with a flatter tail, reflecting the run-up of late recoveries against realised defaults.
- A vintage whose curve is materially above peers at the same MOB should be investigated by origination channel, geography, vehicle type, borrower segment, pricing and underwriting characteristics.

The portfolio under review delivered a current pool factor of 0.58 — consistent with a book that has amortised through scheduled principal, prepayments and a small tail of defaults/repossessions.

10. IFRS 9 ECL Framework

The IFRS 9 framework translates credit deterioration into expected credit loss through three regulatory stages:



- **Stage 1** — Performing loans; 12-month ECL recognised.
- **Stage 2** — Significant increase in credit risk (SICR); lifetime ECL recognised.
- **Stage 3** — Credit-impaired; lifetime ECL with non-accrual treatment.

ECL IDENTITY

$$\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Stage migration therefore matters: deterioration increases expected loss before defaults are fully realised.

The Credit Risk & IFRS 9 Analytics page summarises the pool at regulatory-grain. The IFRS 9 ECL Analysis page breaks the calculation down stage-by-stage.

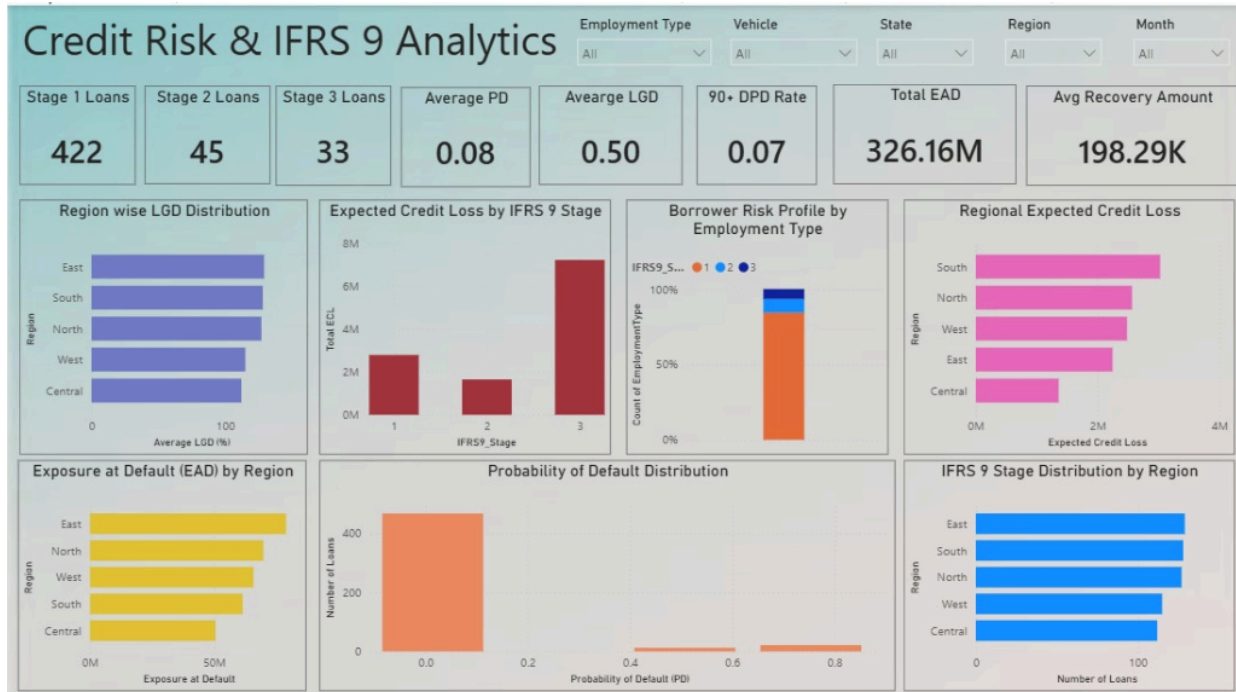
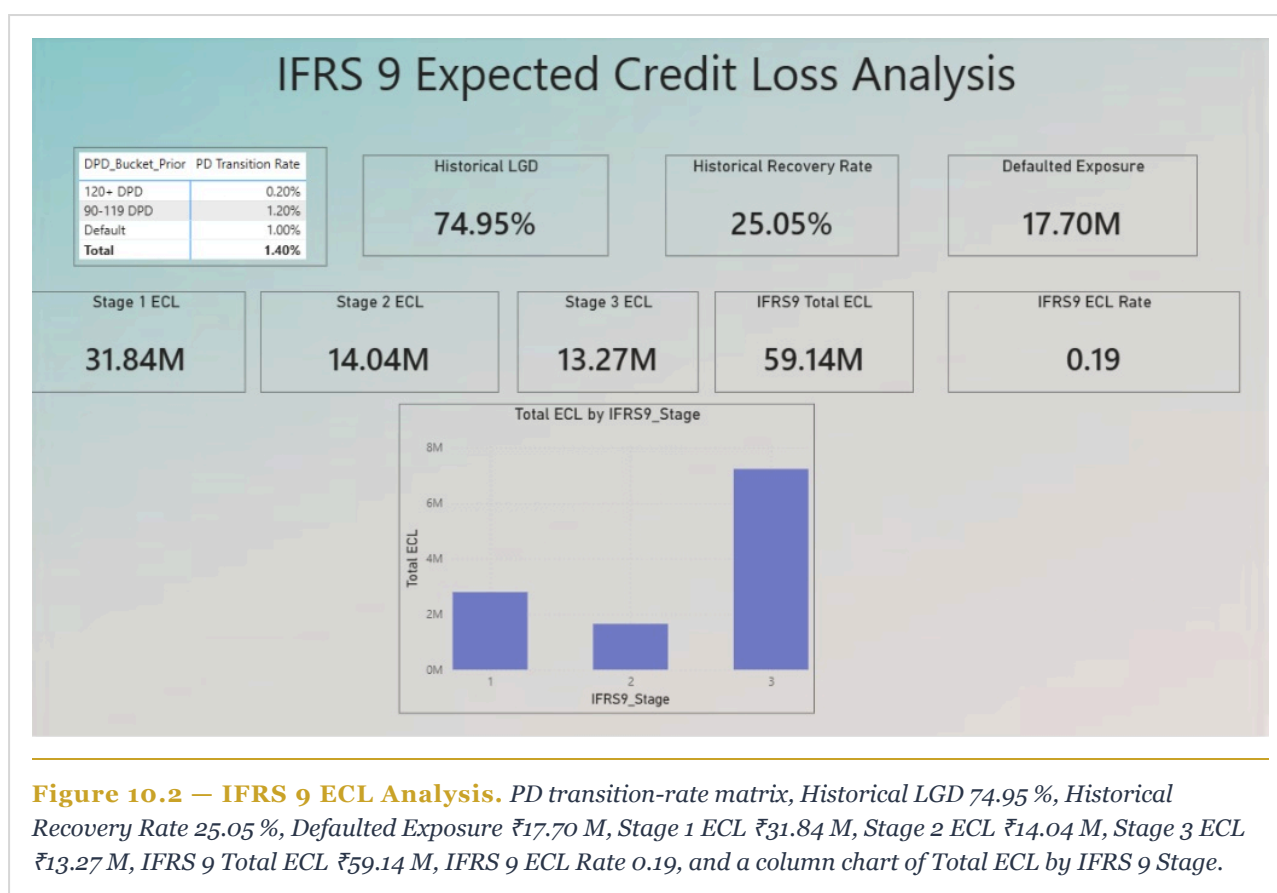


Figure 10.1 — Credit Risk & IFRS 9 Analytics. Regional LGD distribution; Expected Credit Loss by IFRS 9 Stage; Borrower Risk Profile by Employment Type; Regional Expected Credit Loss; Exposure at Default (EAD) by Region; Probability of Default Distribution; IFRS 9 Stage Distribution by Region. KPIs: 422 Stage 1 loans, 45 Stage 2, 33 Stage 3; Avg PD 0.08; Avg LGD 0.50; 90+ DPD rate 0.07; Total EAD ₹326.16 M; Avg Recovery Amount ₹198.29 K.



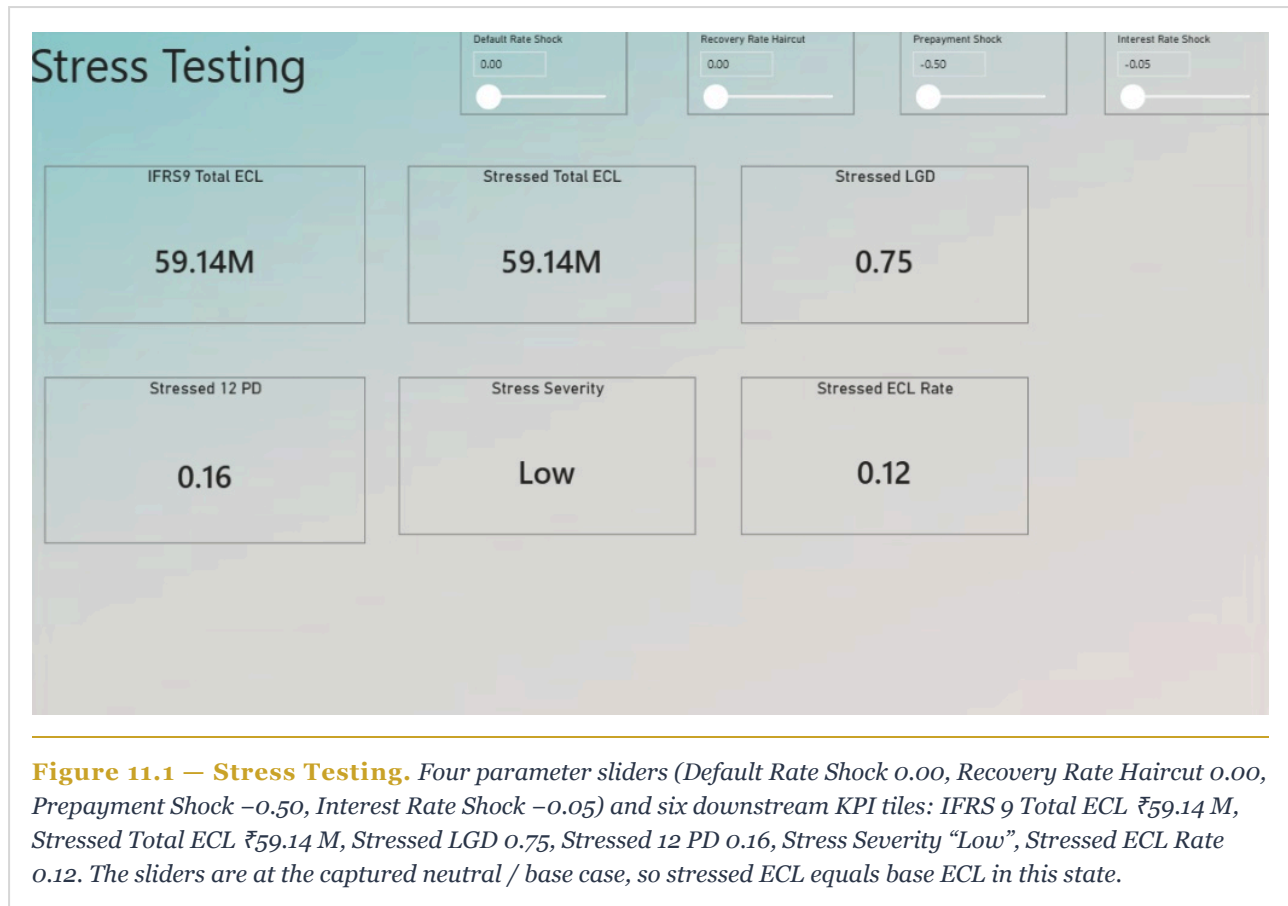
Reconciliation of ECL figures across pages

The Executive Overview tile reports a **net / overview ECL of ₹11.66 M**. The IFRS 9 ECL Analysis page reports a **lifetime-stage-clause ECL of ₹59.14 M**. Both figures are correct within their page-specific analytical grain: the overview represents an ECL snapshot integrated into the net-loss / outstanding balance decomposition, while the IFRS 9 page represents the full stage-1, stage-2, stage-3 lifetime-ECL computation. They are not alternate answers to the same question; they answer different questions about the same portfolio. Cross-page reconciliation tabs should be added in a production version of this dashboard.

This is an academic implementation of the IFRS 9 methodology. Formal IFRS 9 use requires institution-specific policies, forward-looking macroeconomic scenarios, discounting assumptions and full model governance.

11. Stress Testing & What-If Analysis

The Stress Testing page closes the chain from historical observation to forward-looking decision support. Four what-if parameters modify the underlying ECL measures through disconnected slicers and parameter measures.



Stress dimensions and propagation

- **Default Rate Shock** — multiplicative increase in the PD curve.
- **Recovery Rate Haircut** — multiplicative decrease in the recovery curve (raises LGD).
- **Prepayment Shock** — modifies the prepayment-speed assumption, affecting pool run-off and credit-enhancement retention.
- **Interest Rate Shock** — modifies the discount rate used in lifetime ECL.

Default Rate ↑ + Recovery Rate ↓ + Funding / Rate Pressure ↑



PD ↑ · LGD ↑ · EAD remains material



Expected Loss ↑

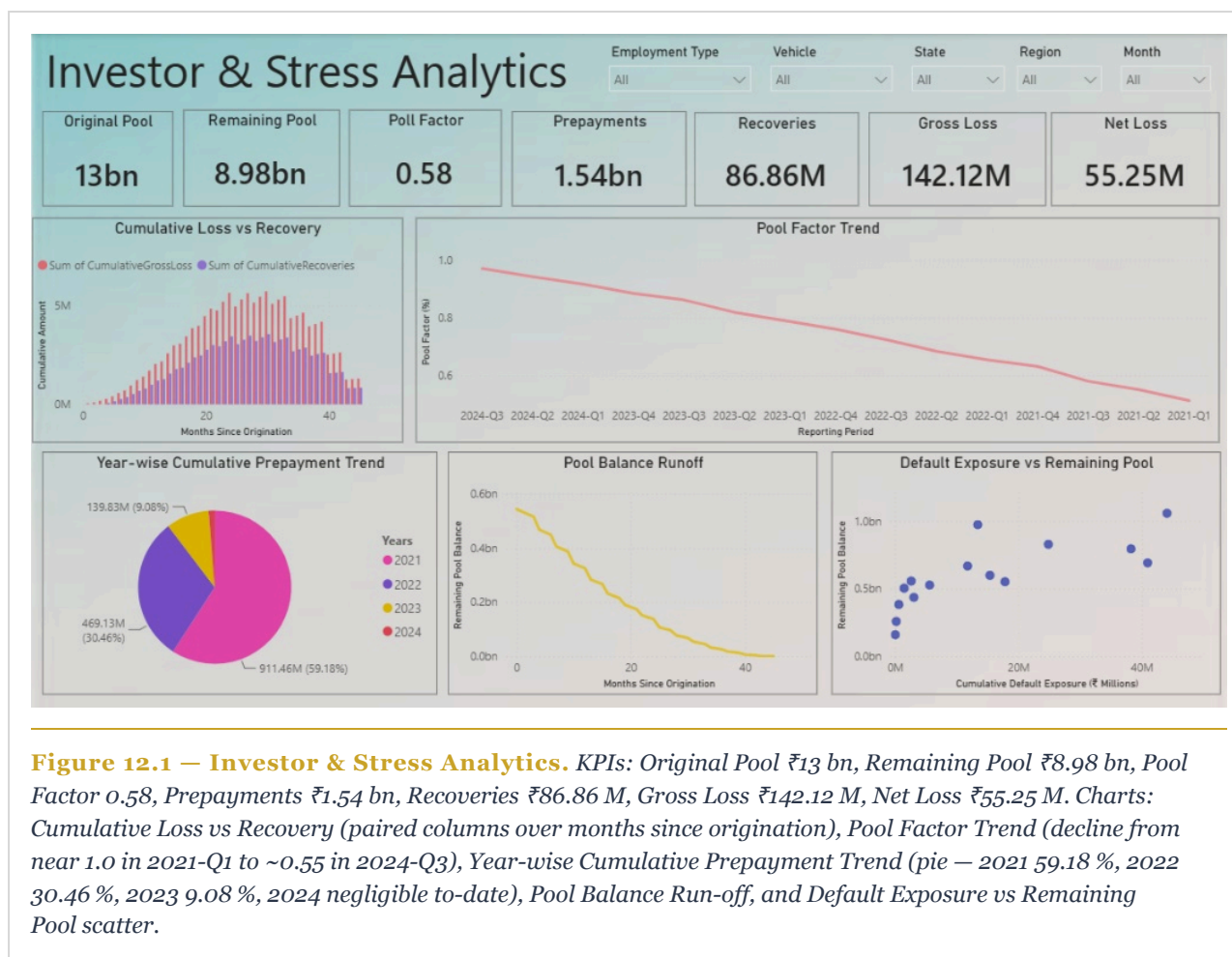


Investor & Pool Risk ↑

The framework therefore allows management to ask: “*What happens to portfolio losses if credit conditions deteriorate beyond the base case?*” A production implementation should define and document calibrated Base, Moderate, Severe and Crisis parameter sets.

12. Investor Reporting

The investor view combines pool-level amortisation, prepayment behaviour, recoveries and losses. Both the Investor & Stress Analytics page and the Executive Overview page expose this layer.



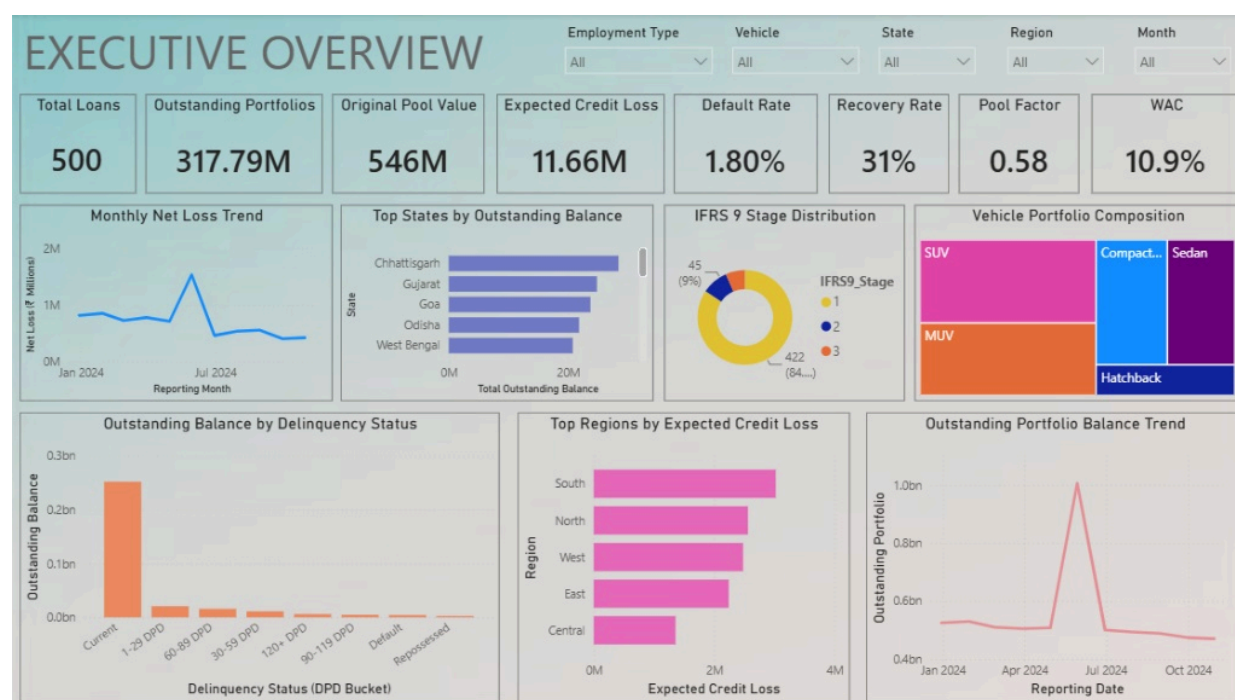


Figure 12.2 — Executive Overview. KPIs: Total Loans 500, Outstanding Portfolios ₹317.79 M, Original Pool Value ₹546 M, Expected Credit Loss ₹11.66 M, Default Rate 1.80 %, Recovery Rate 31 %, Pool Factor 0.58, WAC 10.9 %. Charts: Monthly Net Loss Trend, Top States by Outstanding Balance (Chhattisgarh, Gujarat, Goa, Odisha, West Bengal), IFRS 9 Stage Distribution donut (Stage 1 422 / 84 %, Stage 2 45 / 9 %, Stage 3 33 / 7 %), Vehicle Portfolio Composition treemap (SUV, MUV, Compact, Sedan, Hatchback), Outstanding Balance by Delinquency Status, Top Regions by Expected Credit Loss, Outstanding Portfolio Balance Trend.

Layers an investor-grade report should reconcile

- **Pool Summary.** Beginning-of-period and end-of-period balance, scheduled principal collected, unscheduled prepayments, losses, recoveries and pool factor.
- **Delinquency.** Migration between buckets, cure rate, roll-to-default rate.
- **Loss & recovery.** Cumulative gross and net losses, recovery rate, average time to recovery.
- **Prepayment.** Monthly prepayment amount, SMM, annualised CPR.
- **Waterfall.** Sequential allocation of collections to senior tranche, mezzanine tranche and equity residual.
- **Trigger status.** OC tests, IC tests, delinquency triggers, cumulative-loss triggers.

13. Portfolio Performance

The Portfolio Performance page connects operational cash collection with credit deterioration. Billing versus collections provides a cash-flow control view; collection efficiency translates realised payments to scheduled amounts; default-vs-recovery and gross-vs-net loss bridge asset quality to investor loss outcomes; the pool-factor trend provides a seasoning and amortisation view.

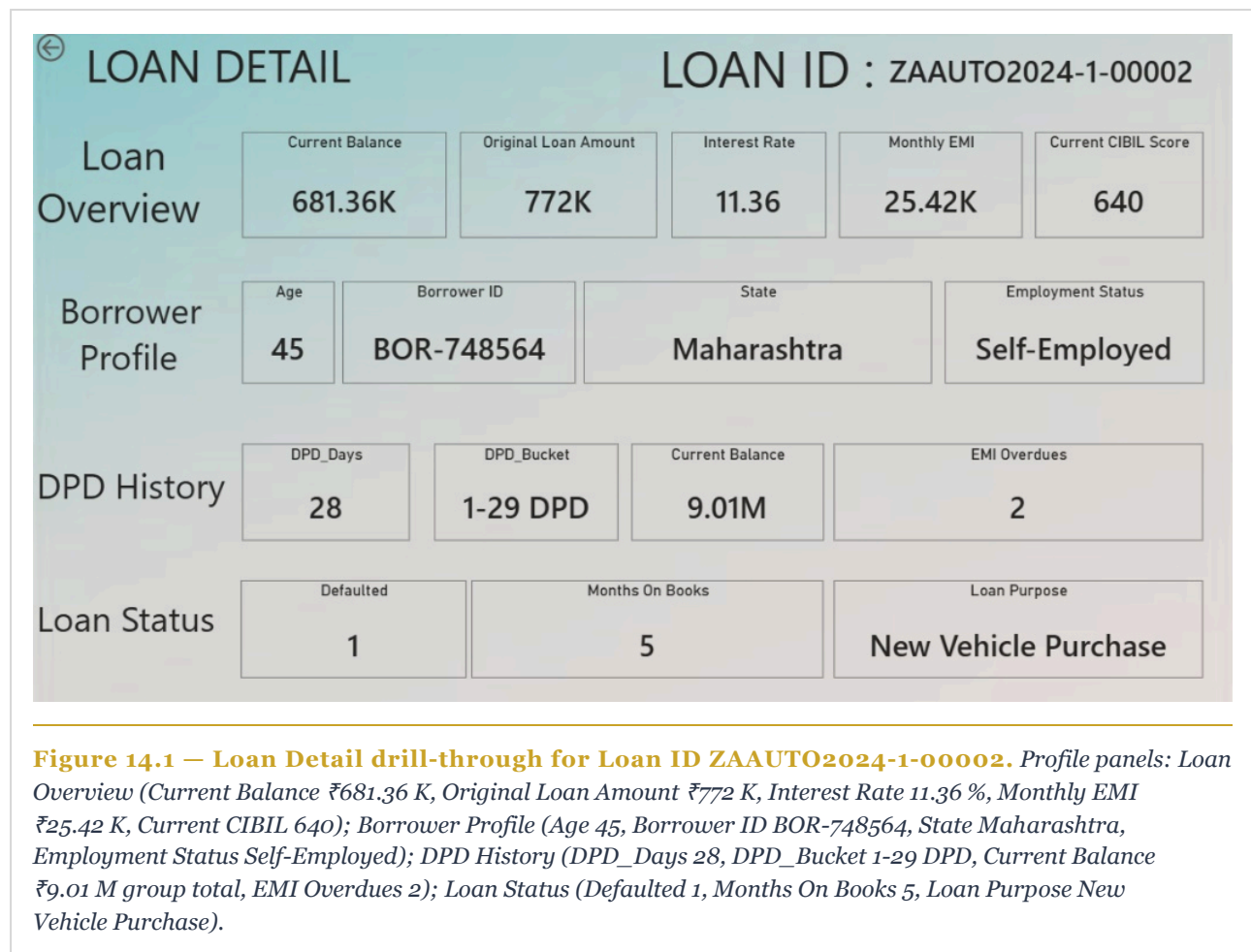


Figure 13.1 — Portfolio Performance. KPIs: Collection Rate 0.86, Average DPD 21.21, Maximum DPD 694, Total Collections ₹224.31 M, Collection Efficiency 1.29, Pool Factor 0.68, Average DTI 0.19. Charts: Monthly Billing vs Collections (column / line over 2024), Collection Efficiency Trend (monthly volatility, mean ~1.3), Regional Collection Efficiency vs Default Rate (Central, East, North, South, West), Default vs Recovery Trend, Pool Factor Trend, Gross Loss vs Net Loss Trend.

Several operational signals sit alongside the credit signals. Collection efficiency above 1.0 reflects partial pre-payment of future EMIs; max DPD of 694 indicates the longest-aged delinquent exposure; average DPD of 21 masks the skew from the chronic tail.

14. Loan-Level Drill-Through

The Loan Detail page moves from a portfolio-level signal to a single borrower record. It performs the role of operational investigation when a high-risk segment, region, or cohort anomaly is detected at the aggregate level.



Why drill-through matters in a risk framework

- The portfolio-level 30+ DPD rate moves slowly; loan-level drill-through isolates the immediate operating cause (collections, restructuring, repossession).
- A risk analyst can move from a high-risk region or employment segment to contributing loans and inspect CIBIL migration, DPD history and EMI overdues.
- The drill-through is also the bridge into remediation workflows: the Self-Employed × Maharashtra × 1-29 DPD profile in this pool has direct implications for collection-script design.

15. Case Study 1 — The 2008 Global Financial Crisis

15.1 Background

The 2007 – 2010 financial crisis is the canonical case study for securitisation analytics failure. The U.S. subprime-mortgage-backed-securities market expanded rapidly through securitisation; the deterioration of underlying mortgage performance, falling house prices and downgrades of structured products together produced broad financial market stress. The Federal Reserve and FDIC record the crisis as spanning roughly 2007 – 2010, with the National Bureau of Economic Research dating the recession December 2007 – June 2009. *(Source: Federal Reserve History — Subprime Mortgage Crisis; FDIC — Origins of the Crisis)*

15.2 The credit deterioration chain



This maps directly to our analytical architecture. A July 2007 S&P action placed 612 subprime RMBS on credit watch — explicitly cluster-level early-warning on the underlying mortgage-pool delinquency distribution. *(Source: Pace Law Library Financial Crisis Timeline)*

15.3 DPD & roll-rate perspective

The early signal in any auto-loan or mortgage portfolio is movement between delinquency states. Our roll-rate matrix measures migration between Current, 1-29, 30-59, 60-89, 90-119, Default and Repossessed. A rising Current → 1-29 transition indicates early deterioration *before* default counts begin to dominate portfolio-level loss metrics; a rising 60-89 → 90-119 transition is the penultimate warning before default.

15.4 Vintage perspective

Vintage curves compare origination cohorts at the same Months-on-Book. A 2006-vintage subprime mortgage would have shown a steeper cumulative default curve at 24 M than a 2003-vintage. The same analytical lens applies to our 2021-Q1 — 2024-Q3 vintages on the Vintage / Static-Pool Analysis page.

15.5 IFRS 9 ECL perspective

Stage 1 loans → Stage 2 → Stage 3



12-month ECL → Lifetime ECL → Lifetime impaired-ECL



Increasing ECL provisioning requirement

15.6 Stress-testing perspective

The what-if framework simulates exactly the mechanism that was observed: rising defaults and deteriorating recoveries drive PD ↑ and LGD ↑ while EAD remains material, so stressed ECL rises. Lehman Brothers' 2008 collapse and the broader credit-market freeze are the historical analogue of “Stress Severity = Severe”.

15.7 Dashboard indicators that would have fired

Indicator	Risk signal
DPD roll-rate matrix	Increasing delinquency migration
90+ DPD rate	Severe delinquency
Default rate	Increasing credit losses
Vintage cumulative loss	Weak origination cohorts
LGD / Recovery rate	Increasing loss severity
Stage 2 exposure	Significant increase in credit risk
Stage 3 exposure	Credit-impaired / defaulted
IFRS 9 ECL	Increasing expected losses
Pool factor	Remaining collateral exposure
Stressed ECL	Vulnerability under adverse assumptions

15.8 Key lesson for securitisation analytics

Securitisation risk cannot be assessed using a single default-rate metric. The integrated chain is: loan quality → delinquency migration → defaults → recoveries / LGD → vintage performance → ECL → stress scenarios → pool / investor risk. This is what the dashboard built across Days 7 – 10 measures.

16. Case Study 2 — The IL&FS Crisis

16.1 Background

Infrastructure Leasing & Financial Services (IL&FS) was a large Indian infrastructure-finance group with a wide network of subsidiaries. SEBI disclosures document the consolidated borrowings of IL&FS Group at ₹91,091 crore as at 31 March 2018. The crisis became acute in early September 2018 when IL&FS defaulted on a series of obligations including commercial paper. The Government of India moved to supersede the existing board and the National Company Law Tribunal appointed a new board on 1 October 2018; subsequent restructuring classified group entities into Green, Amber and Red categories based on a 12-month cash-flow solvency test. *(Source: Firstpost IL&FS coverage, September 2018; IL&FS India disclosure — Green/Amber/Red categorisation, 3 April 2019; NCLT order 1 October 2018)*

16.2 The risk chain that developed



16.3 Liquidity amplifies credit risk

Once market confidence weakened, lower confidence fed into higher funding cost, reduced access to funding, greater refinancing pressure and ultimately higher probability of default — a feedback loop where credit risk and liquidity risk reinforce each other. RBI specifically noted that the September 2018 default affected market confidence and produced liquidity stress and higher borrowing costs for non-banking financial companies. *(Source: RBI communications on NBFC liquidity, 2018)*

16.4 Connection to our DPD & credit-risk model

IL&FS risk concept	Our analytical equivalent
Payment stress	DPD analysis
Deteriorating borrowers	Roll-rate migration
Defaults	Default rate
Loss severity	LGD / Recovery rate
Exposure at risk	EAD
Expected losses	IFRS 9 ECL
Stress scenario	What-if stress testing
Remaining collateral / pool	Pool factor
Funding / liquidity pressure	Scenario analysis

16.5 Early-warning perspective

Default is not the first warning signal. Our DPD framework provides earlier signals — transition from Current → 1-29 → 30-59 → 60-89 → 90-119 → 120+ → Default — and our roll-rate matrix tells us whether borrowers are increasingly migrating toward worse states.

16.6 IFRS 9 perspective

ECL IDENTITY IN AN IL&FS-STYLE STRESS SCENARIO

A deterioration in credit quality can affect PD, LGD and EAD simultaneously.

PD ↑ — probability of default rises.

LGD ↑ — recovery assumptions deteriorate.

EAD — exposure remains substantial on undrawn / non-amortising facilities.

Therefore: ECL ↑ before the entire portfolio reaches default.

16.7 Stress-testing perspective

Default Rate ↑ · Recovery Rate ↓ · Funding / Rate Pressure ↑



Expected Loss ↑



Investor Risk ↑

16.8 Investor perspective

Strong investor reporting must combine credit risk, liquidity risk, funding risk, concentration risk and refinancing risk. The IL&FS episode illustrates how stress at one highly interconnected group can propagate through the broader financial system — relevant to any securitisation pool with material indirect exposure via mutual funds, banks, pension funds or NBFC counterparties.

16.9 Regulatory response

Following the crisis, the government superseded the IL&FS board; the NCLT appointed a new board on 1 October 2018. Subsequent restructuring classified group entities into Green, Amber and Red categories based on a 12-month cash-flow solvency assessment — a forward-looking framework rather than a backward-looking accounting view. (Source: NCLT order 1 October 2018; IL&FS India disclosure 3 April 2019)

16.10 What our dashboard would monitor

- **Credit deterioration.** DPD → roll-rate → default.
- **Expected losses.** PD → LGD → EAD → ECL.
- **Portfolio deterioration.** Vintage curves → cumulative loss.
- **Stress.** Default shock → recovery haircut → stress ECL.
- **Investor position.** Pool balance → losses → recoveries → risk status.

16.11 Key lessons

Lesson 1 — Liquidity amplifies credit risk. A borrower may begin with an asset-quality problem that is then converted into a default through inability to refinance.

Lesson 2 — Early-warning indicators matter. Monitoring DPD migration and roll-rates is more useful than waiting for realised defaults.

Lesson 3 — Concentration matters. Large exposures to interconnected entities can transmit stress across the system.

Lesson 4 — Stress testing must be forward-looking. Base-case performance alone is not enough; the what-if framework applies adverse assumptions before losses materialise.

Lesson 5 — Investor reporting must combine multiple risk dimensions. A strong dashboard answers not only “how much has defaulted?” but also “how much is exposed, how fast is quality deteriorating, how much can be recovered, and how does the pool behave under stress?”.

Note: the case study connects the analytical principles; it does not reproduce the IL&FS balance sheet from the 500-loan auto dataset used in the dashboard.

17. Key Business & Risk Insights

Across ten dashboards, four workstreams (portfolio health, delinquency migration, vintage, IFRS 9 ECL) and a what-if stress layer, the framework delivers several integrated insights.

1. The integrated risk chain. Portfolio composition → delinquency migration → defaults and losses → recovery severity → expected credit loss → stressed ECL → investor exposure. A reductionist reading of default rate alone will materially understate the leading-edge signals that the dashboard exposes.

2. DPD migration is a leading indicator. The roll-rate matrix from one bucket to the next typically leads realised defaults by one to three reporting months. Monitoring the leading edge (Current → 1-29, 1-29 → 30-59) is materially more useful than tracking 90+ DPD, which is a lagging metric.

3. Vintage spread is the cohort-quality signal. Comparing quarterly origination cohorts at matched Months-on-Book separates cohort effects from pool ageing. A defective cohort is then isolated for further investigation by underwriting channel, geography, vehicle type, employment segment and pricing.

4. ECL composition is the regulatory anchor. The dashboard distinguishes Stage 1, Stage 2 and Stage 3 ECL so that provisioning density can be tracked. Recovery rate of 25.05 % and historical LGD of 74.95 % (from the IFRS 9 ECL Analysis page) anchor the sensitivity of total ECL to recovery-rate shocks.

5. Stress vulnerability is parameter-visible. All four what-if sliders (default shock, recovery haircut, prepayment shock, interest-rate shock) write through to the stressed KPI tiles. The current neutral state shows Stress Severity = Low and Stressed ECL Rate of 0.12. Production calibration would document scenario severity bands.

6. Investor reporting requires reconciliation. The Executive Overview tile of ₹11.66 M and the stage-clause lifetime ECL of ₹59.14 M are computed at different analytical grains; production reporting must reconcile them with explicit methodology notes.

7. Multi-risk monitoring is necessary. Both case studies — the 2008 GFC and the 2018 IL&FS episode — show that credit deterioration, liquidity stress, refinancing exposure and concentration combine to drive the ultimate loss. A dashboard that monitors credit risk alone is insufficient.

8. Cross-crisis structure. In both 2008 and IL&FS, the decrease in investor confidence (and the rise in funding cost) is the amplificatory mechanism by which base-case credit deterioration translates into realised investor loss. This is precisely the dimension captured by the Stress Testing page's Recovery-Rate Haircut and Interest-Rate Shock parameters.

18. Limitations & Assumptions

- **Illustrative dataset.** The source data is a 500-loan academic / demonstration dataset. It is not an audited institutional portfolio and is not suitable for production regulatory reporting.
- **Simplified single-product ABS.** The dataset covers a single asset class (auto loans) and a single originator — cross-collateral / multi-originator concentration logic is not in scope.
- **Simple staging.** The IFRS 9 implementation uses the standard Stage 1 / 2 / 3 staging rules with 12-month and lifetime ECL separately. It does not incorporate forward-looking macroeconomic scenarios, multiple probability-weighted scenarios, or discounting refinements required by the full standard.
- **Static loan-level model.** The economic logic (servicing costs, recovery lag, prepayment seasonality) is simplified; production implementations require institution-specific calibration and back-testing.
- **Scenario parameters.** Default-state stress severity is Low (Stressed ECL Rate 0.12). A calibrated Base / Moderate / Severe / Crisis parameter set would be required for production use.
- **Investor reporting.** The rating-agency data-tape fields are illustrative; production data tapes require transaction-specific field mapping to S&P, Moody's, CRISIL and ICRA schemas.
- **Waterfall and trigger tests.** Sequential tranche allocation and OC / IC trigger tests are not implemented in this academic version.
- **Governance gap.** Row-level security for five or more roles, DAX Studio performance testing, Excel ECL cross-checks and full waterfall validation are referenced in the manual but are not included in this prototype.
- **Cross-page reconciliation.** Two ECL figures (Executive Overview ₹11.66 M vs IFRS 9 lifetime ₹59.14 M) are produced at different analytical grains; production reporting would include a reconciliation tab.

19. Conclusion

This project delivers a consolidated Power BI framework that moves beyond basic reporting into structured credit-risk analytics. The dimensional model supports reusable DAX across an executive view, a risk view, a vintage view and an investor view. Each dashboard page is anchored to a single business question; cross-page consistency is enforced through shared measures and shared grain definitions.

The strongest contribution of the framework is the integration of historical and forward-looking perspectives. DPD and roll-rates identify current deterioration; vintage analysis reveals cohort behaviour; IFRS 9 ECL translates credit-risk assumptions into expected loss; what-if stress testing evaluates resilience. Together, these layers match the Day 7 – Day 11 workflow required by the project manual.

The two case studies reinforce the business rationale. The 2008 subprime crisis shows how granular mortgage deterioration propagated through securitised assets and investor markets; the IL&FS episode shows how leverage, refinancing and liquidity can amplify credit stress. In both cases, the same integrated analytical lens — delinquency migration, vintage comparison, expected credit loss, stress testing — is what a strong investor-grade dashboard must deliver.

As a final deliverable, the project functions as a strong analytical prototype for securitisation risk monitoring. Formal validation, governance, RLS security, transaction-specific waterfall logic, and full regulatory mapping would be the natural next steps toward production use.

20. Appendix A — Important DAX Measures

This appendix consolidates the DAX formulas referenced throughout the report. The measures are grouped by layer. All measures are formula-driven; no hard-coded values are used.

Pool & weighted metrics

POOL FACTOR

```
Pool Factor = DIVIDE( SUM ( Loans[CurrentBalance] ), CALCULATE ( SUM (
Loans[OriginalLoanAmount] ), ALL ( DimDate
) ) )
```

WEIGHTED AVERAGE COUPON (WAC)

```
WAC = DIVIDE ( SUMX ( Loans, Loans[InterestRate] * Loans[CurrentBalance] ), SUM
( Loans[CurrentBalance] ) )
```

WEIGHTED AVERAGE MATURITY (WAM)

```
WAM = DIVIDE ( SUMX ( Loans, Loans[RemainingTerm] * Loans[CurrentBalance] ),
SUM ( Loans[CurrentBalance] ) )
```

WEIGHTED AVERAGE LOAN AGE (WALA)

```
WALA = DIVIDE ( SUMX ( Loans, Loans[MonthsOnBook] * Loans[CurrentBalance] ),
SUM ( Loans[CurrentBalance] ) )
```

Credit-quality measures

DELINQUENCY RATE (30+)

```
Delinquency Rate = DIVIDE ( CALCULATE ( SUM ( Loans[CurrentBalance] ),
Loans[DelinquencyDays] >= 30 ), SUM (
Loans[CurrentBalance] ) )
```

SERIOUS DELINQUENCY RATE (60+)

```
Serious Delinquency Rate = DIVIDE ( CALCULATE ( SUM ( Loans[CurrentBalance] ),
Loans[DelinquencyDays] >= 60 ),
SUM ( Loans[CurrentBalance] ) )
```

90+ DPD RATE (NPA)

```
90Plus DPD Rate = DIVIDE ( CALCULATE ( SUM ( Loans[CurrentBalance] ),
Loans[DelinquencyDays] >= 90 ), SUM (
Loans[CurrentBalance] ) )
```

Prepayment, default & loss

SINGLE MONTH MORTALITY (SMM)

```
SMM = DIVIDE ( SUM ( Loans[PrepaymentAmount] ), CALCULATE ( SUM (
Loans[CurrentBalance] ), PREVIOUSMONTH (
DimDate[Date] ) ) - SUM ( Loans[ScheduledPrincipal] ) )
```

CONDITIONAL PREPAYMENT RATE (CPR)

```
CPR = 1 - POWER ( 1 - [SMM], 12 )
```

CONSTANT DEFAULT RATE (CDR)

```
CDR = VAR MDR = DIVIDE ( [Defaults This Month], CALCULATE ( SUM (
Loans[CurrentBalance] ), PREVIOUSMONTH (
DimDate[Date] ) ) ) RETURN 1 - POWER ( 1 - MDR, 12 )
```

COLLECTION EFFICIENCY

```
Collection Efficiency = DIVIDE ( SUM ( Loans[ActualPayment] ), SUM (
Loans[ScheduledPaymentDue] ) )
```

RECOVERY RATE

```
Recovery Rate = DIVIDE ( CALCULATE ( SUM ( DynamicLoss[Recoveries_ThisMonth] )
), CALCULATE ( SUM (
DynamicLoss[GrossLoss_ThisMonth] ) ) )
```

Roll-rate (DPD transition)

ROLL RATE (BUCKET → BUCKET)

```
Roll Rate = VAR FromBucket = SELECTEDVALUE ( RollRateMatrix[FromBucket] ) VAR
ToBucket = SELECTEDVALUE (
    RollRateMatrix[ToBucket] ) VAR CurrentDate = MAX ( DimDate[Date] ) VAR
PriorDate = EDATE ( CurrentDate, -1 ) VAR
BalanceInFrom = CALCULATE ( SUM ( FactLoanPerformance[CurrentBalance] ),
FactLoanPerformance[DPD_Bucket] =
    FromBucket, DimDate[Date] = PriorDate ) VAR BalanceTo = CALCULATE ( SUM (
FactLoanPerformance[CurrentBalance] ),
    FactLoanPerformance[DPD_Bucket] = ToBucket, DimDate[Date] = CurrentDate )
RETURN DIVIDE ( BalanceTo, BalanceInFrom
    )
```

Vintage / static-pool

STATIC CUMULATIVE NET LOSS RATE

```
Static CNL Rate = VAR SelectedVintage = SELECTEDVALUE ( StaticPool[VintageID] )
VAR MaxMOB = MAX (
    StaticPool[MonthsOnBook] ) VAR CumNetLoss = CALCULATE ( MAX (
StaticPool[CumulativeNetLoss] ),
    StaticPool[VintageID] = SelectedVintage, StaticPool[MonthsOnBook] = MaxMOB )
VAR OrigBalance = CALCULATE ( MAX (
    StaticPool[OriginalPoolBalance] ), StaticPool[VintageID] = SelectedVintage )
RETURN DIVIDE ( CumNetLoss,
    OrigBalance )
```

MARGINAL MONTHLY LOSS RATE

```
Marginal Monthly Loss Rate = VAR CurrentMOB = MAX ( StaticPool[MonthsOnBook] )
VAR CurrentCNL = [Static CNL Rate]
VAR PriorCNL = CALCULATE ( [Static CNL Rate], StaticPool[MonthsOnBook] =
CurrentMOB - 1 ) RETURN CurrentCNL -
    PriorCNL
```

IFRS 9 ECL measures

STAGE CLASSIFICATION

```
IFRS 9 Stage = SWITCH ( TRUE (), Loans[DelinquencyDays] >= 90, 3,
Loans[DelinquencyDays] >= 30 ||
Loans[SICR_Flag] = TRUE (), 2, 1 )
```

EXPOSURE AT DEFAULT (EAD)

```
EAD = SUMX ( Loans, Loans[CurrentBalance] * ( 1 + Loans[CCF] ) -- Credit-
conversion factor for undrawn )
```

IFRS 9 EXPECTED CREDIT LOSS

```
IFRS 9 ECL = SUMX ( Loans, CALCULATE ( [PD] * [LGD] * Loans[EAD] ) )
```

STRESSED ECL (WHAT-IF DRIVEN)

```
Stressed ECL = SUMX ( Loans, ( [PD] * ( 1 + [Default Rate Shock] ) ) * ( [LGD]
* ( 1 + [Recovery Rate Haircut] ) )
* Loans[EAD] )
```

All measures follow `paged.js` — friendly formatting; in Power BI Desktop they should be entered as single lines. The variables and iterator patterns are described in the Day 3 and Day 9 deliverables of the project manual.

21. Appendix B — Dashboard Pages

This appendix reproduces the ten dashboard pages of the Power BI report. Each page is presented as a single figure with caption.

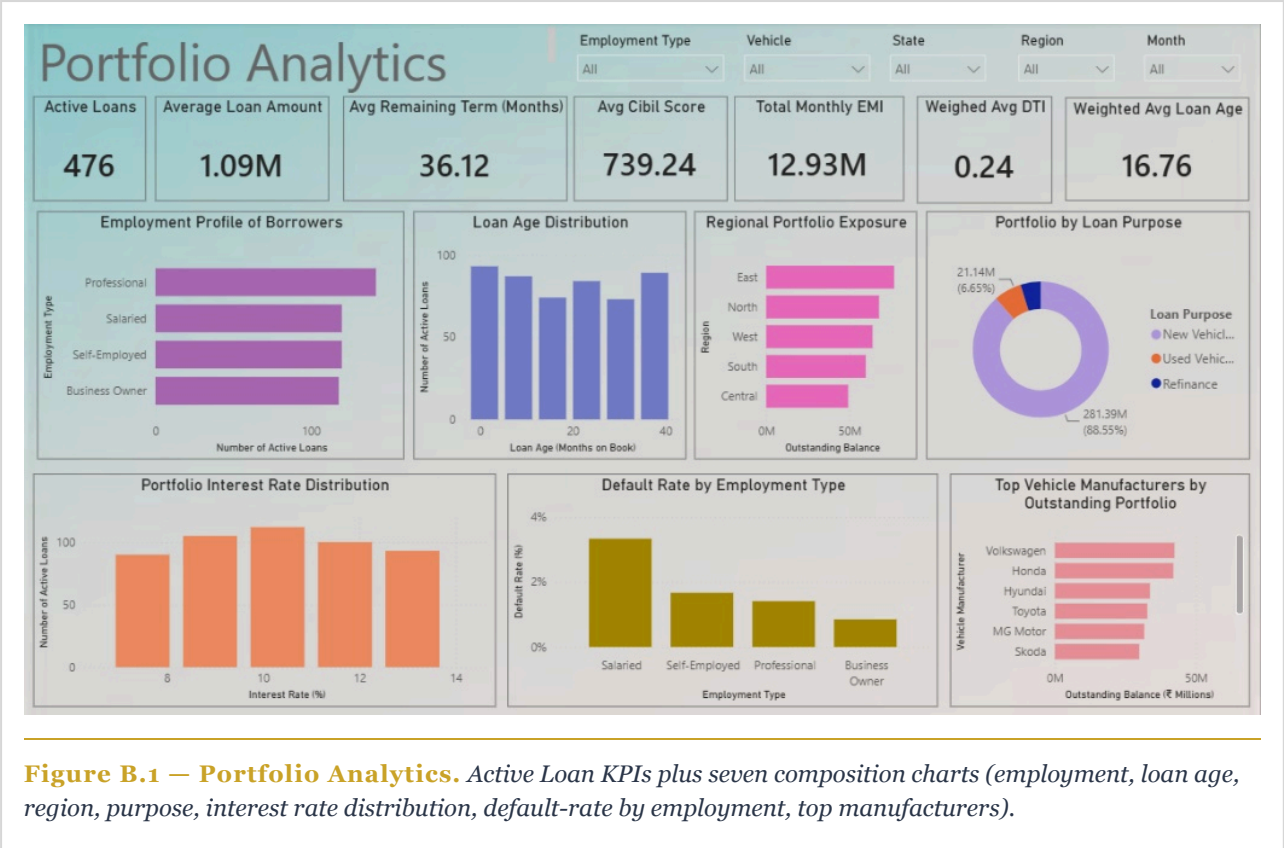


Figure B.1 — Portfolio Analytics. Active Loan KPIs plus seven composition charts (employment, loan age, region, purpose, interest rate distribution, default-rate by employment, top manufacturers).

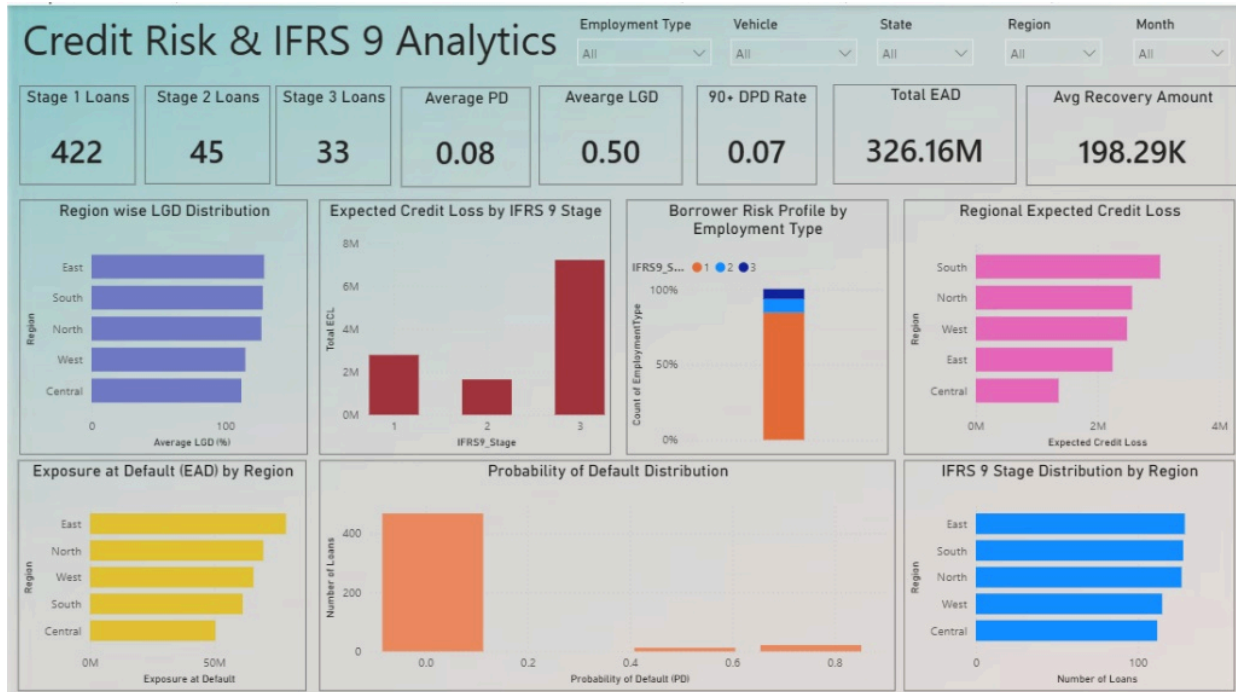


Figure B.2 — Credit Risk & IFRS 9 Analytics. Stage counts (422 / 45 / 33), Avg PD 0.08, Avg LGD 0.50, 90+ DPD 0.07, Total EAD ₹326.16 M, Avg Recovery ₹198.29 K, plus regional LGD, regional ECL, regional EAD, regional stage distribution, PD histogram and stage-wise ECL.

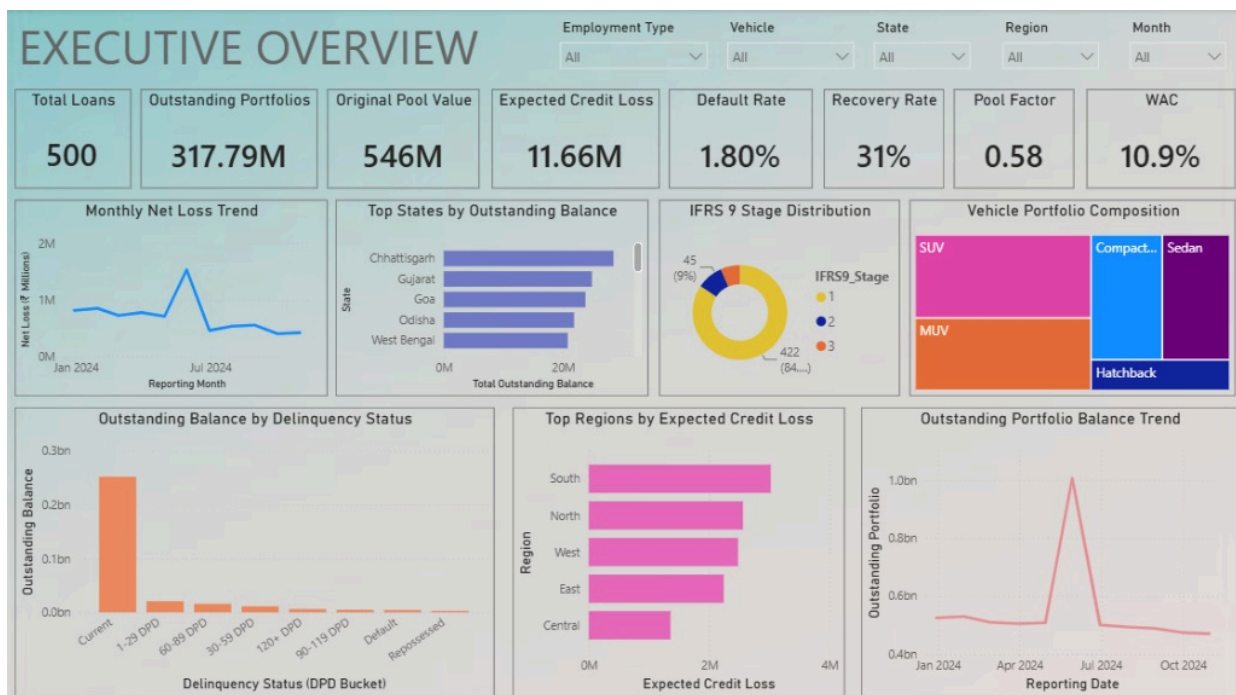


Figure B.3 — Executive Overview. Total Loans 500, Outstanding ₹317.79 M, Original Pool ₹546 M, Expected Credit Loss ₹11.66 M, Default Rate 1.80 %, Recovery Rate 31 %, Pool Factor 0.58, WAC 10.9 %, with monthly net loss trend, top states, IFRS 9 stage donut, vehicle composition treemap and DPD-bucket outstanding balances.



Figure B.4 — Investor & Stress Analytics. Original Pool ₹13 bn, Remaining Pool ₹8.98 bn, Pool Factor 0.58, Prepayments ₹1.54 bn, Recoveries ₹86.86 M, Gross Loss ₹142.12 M, Net Loss ₹55.25 M, plus cumulative-loss-vs-recovery, pool-factor trend, year-wise prepayment pie, run-off curve and default-exposure scatter.



Figure B.5 — Portfolio Performance. Collection Rate 0.86, Avg DPD 21.21, Max DPD 694, Total Collections ₹224.31 M, Collection Efficiency 1.29, Pool Factor 0.68, Avg DTI 0.19, with billing-vs-collections, efficiency trend, regional efficiency vs default, default-vs-recovery, pool factor trend and gross-vs-net loss trend.

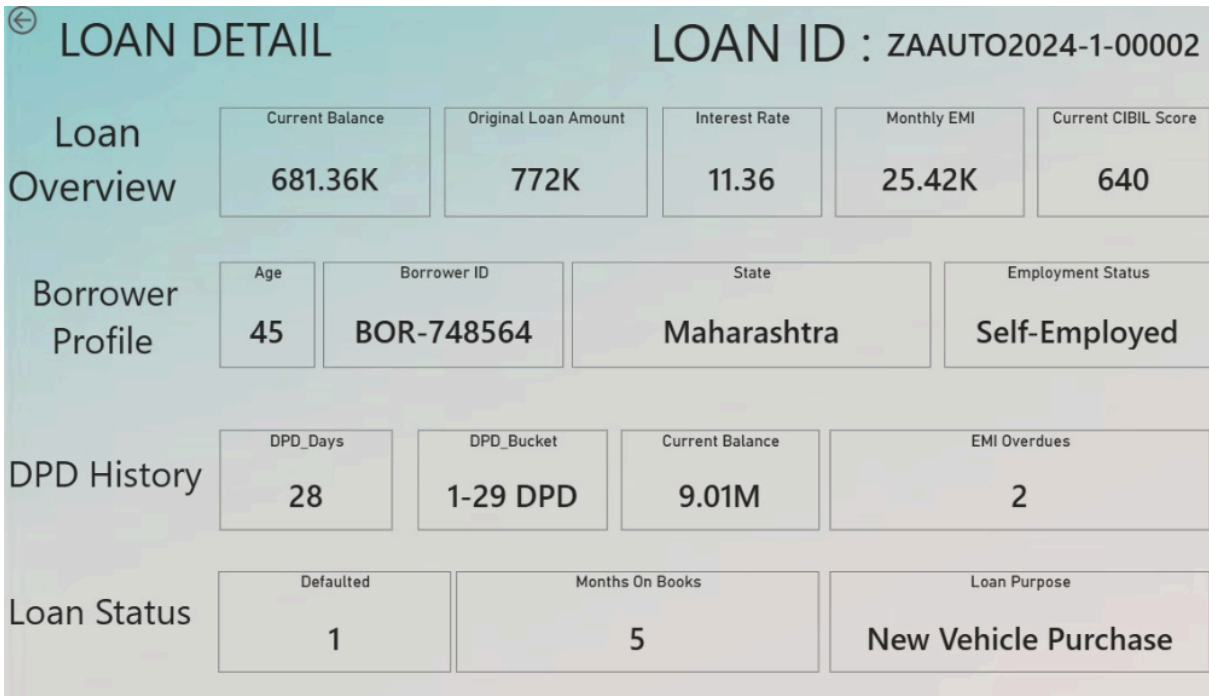


Figure B.6 — Loan Detail. Borrower, balance, DPD and status profile panels for Loan ID ZAAUTO2024-1-00002 (Borrower BOR-748564, Maharashtra, Self-Employed).

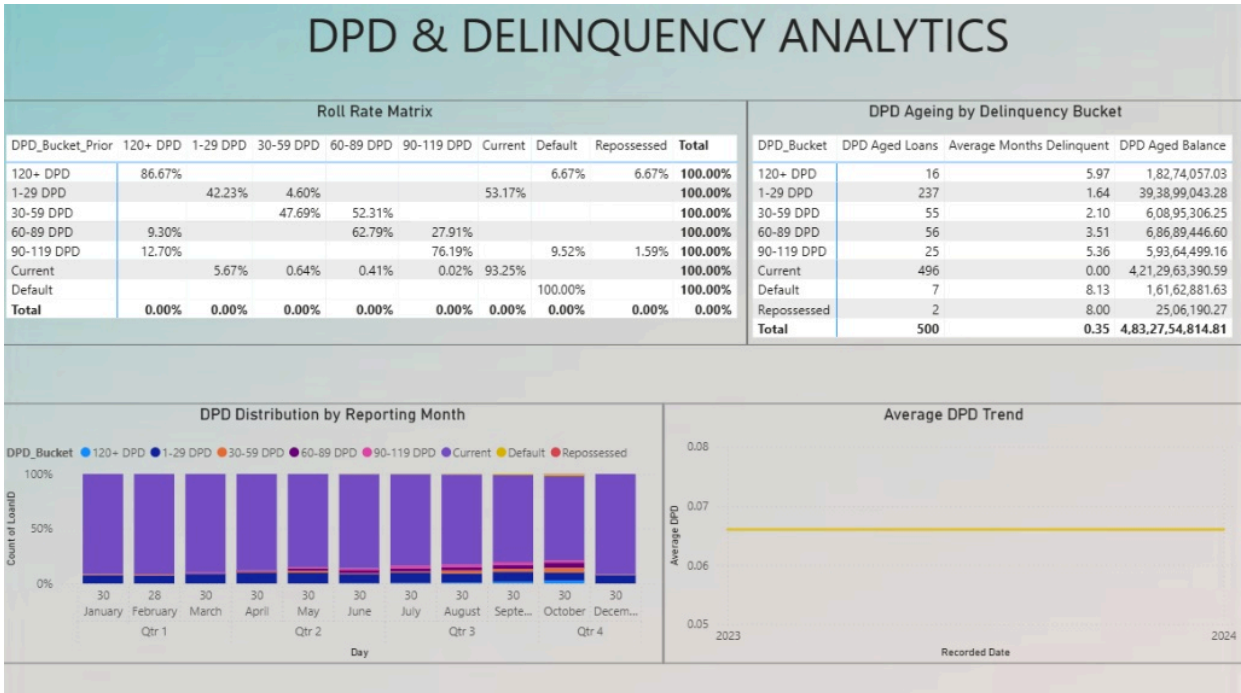


Figure B.7 — DPD & Delinquency Analytics. Roll-rate matrix, DPD ageing table, DPD distribution by reporting month and average DPD trend.

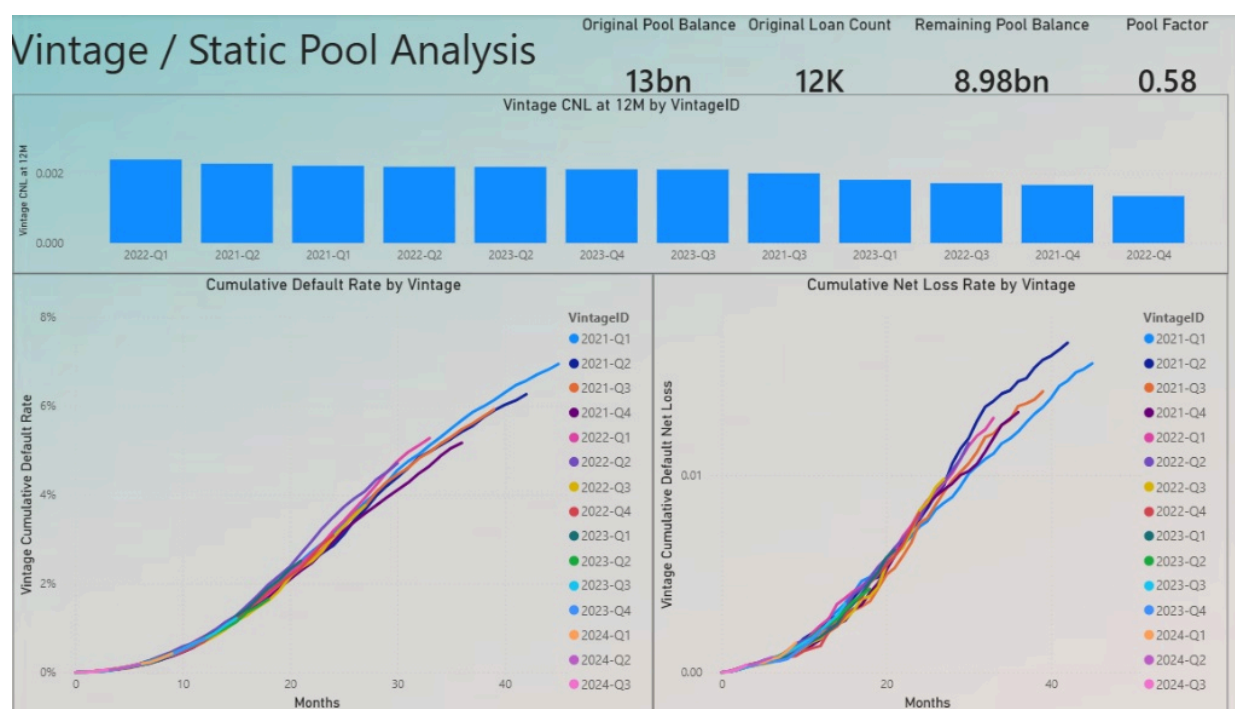


Figure B.8 — Vintage / Static-Pool Analysis. Vintage CNL at 12 M by VintageID, cumulative default rate and cumulative net loss rate by quarterly origination cohort (2021-Q1 — 2024-Q3).

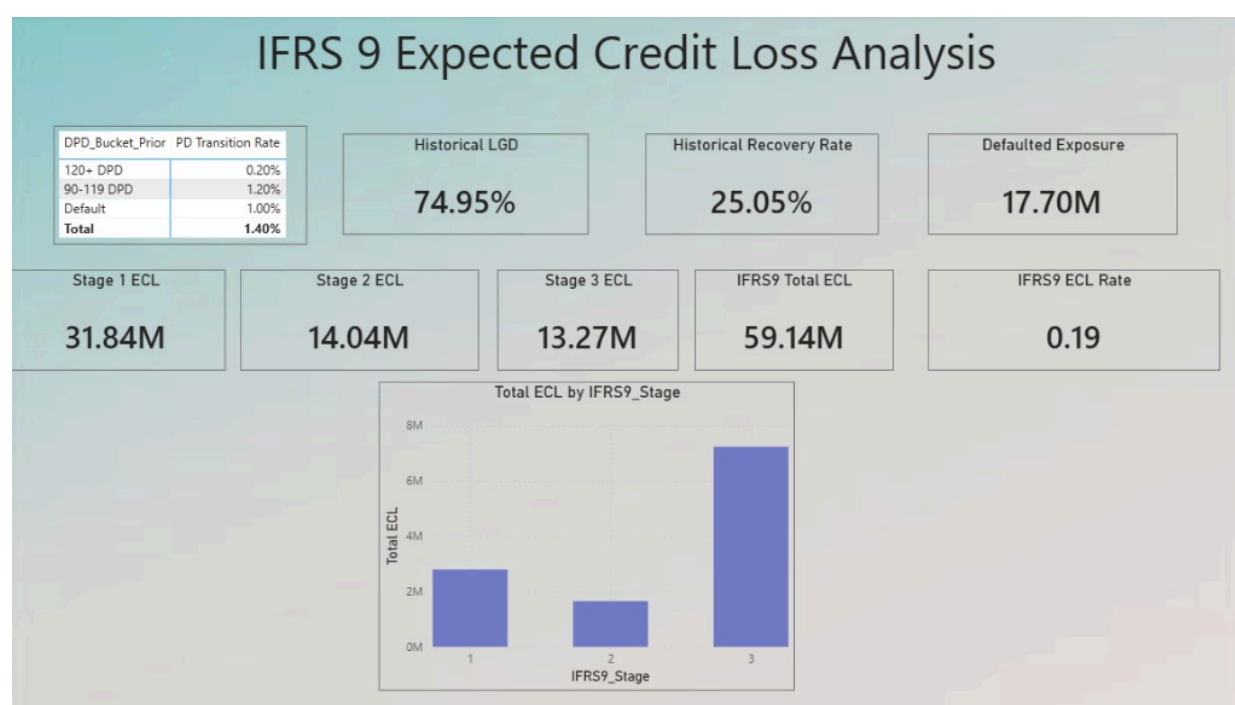


Figure B.9 — IFRS 9 ECL Analysis. PD transition-rate matrix, Historical LGD 74.95 %, Historical Recovery 25.05 %, Defaulted Exposure ₹17.70 M, Stage 1/2/3 ECL ₹31.84 M / ₹14.04 M / ₹13.27 M, Total ECL ₹59.14 M, IFRS 9 ECL Rate 0.19, column chart by stage.

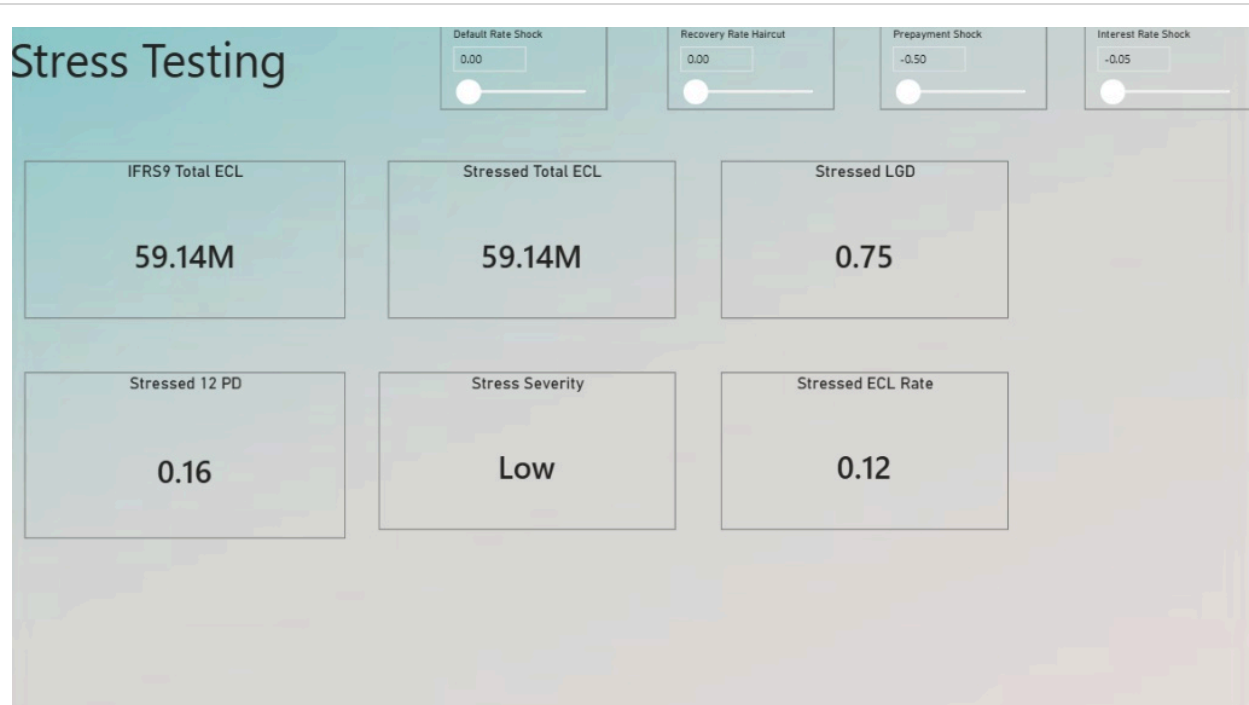


Figure B.10 — Stress Testing. Four parameter sliders (Default Rate Shock 0.00, Recovery Rate Haircut 0.00, Prepayment Shock -0.50, Interest Rate Shock -0.05) and six downstream KPI tiles (IFRS 9 Total ECL ₹59.14 M, Stressed Total ECL ₹59.14 M, Stressed LGD 0.75, Stressed 12 PD 0.16, Stress Severity Low, Stressed ECL Rate 0.12).

All screenshots are reproduced from supplied Power BI captures with original captions preserved; numerical units, percentages and currency formatting are reproduced as captured.